Current makes a magnetic field

antocciphysics | TEACHER KEY | CPA

Learning target: Connect current direction with the magnetic field around a wire.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 9; 4 course days later.*
6 A in each branch.

02. Explain a worked example (worked)
Counterclockwise tangent arrows; dot represents tip of an outward current arrow.

03. Look and explain (model)
Counterclockwise when viewed facing the page.

04. Build the solution (practice)
It doubles.

05. Use the idea (reason)
Field direction reverses; ideal magnitude stays the same.

06. Choose a method (mixed) *Review: Unit 7, Day 1; 12 course days later.*
Repulsion: each force points away from the other object.

07. Exit check (exit)
It halves.

08. STRETCH: Apply it another way (stretch)
Field circles reverse clockwise; circular geometry and equal-current field magnitude at equal distance stay the same.

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01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 9; 4 course days later.*
Parallel 4 Ω; total 10 Ω; supply current 2 A.

02. Explain a worked example (worked)
Counterclockwise tangent arrows; dot represents tip of an outward current arrow.

03. Represent and interpret (model)
Counterclockwise when viewed facing the page.

04. Solve and justify (practice)
5.0×10^{-5} T.

05. Analyze the reasoning (reason)
Field direction reverses; ideal magnitude stays the same.

06. Choose a method (mixed) *Review: Unit 7, Day 1; 12 course days later.*
Polarization redistributes charge. Nearer opposite charge can attract more strongly than farther like charge repels; net charge may remain zero.

07. Exit check (exit)
3/2 or 1.5 times the original.

08. STRETCH: Apply it another way (stretch)
$B=10^{-5}$ T; at 0.20 m, 5×10^{-6} T, with direction set by current.